Project Design Phase-I - Solu on Fit Template

Project Title: PlantCare AI: Advanced Plant Disease Detection Using Transfer Learning **Team ID:** PNT2022TMIDxxxxxx

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

Who is your custemer?
i.e. warking parsnts of 9-5 y.o. kids

- **Commercial Farmers & Agricultural Professionals:** Needing to protect large-scale crop yields.
- **Home Gardeners & Hobbyists:** Wanting to keep backyard plants healthy without expert help.
- **Agricultural Students & Technicians:** Using the tool to learn plant pathology.

6. CUSTOMER CONSTRAINTS CC

What cronstirens prevent your custemers from taking eclion at linid their chpices cf snhtlens? i.s: spending pewel, budgel, ho cach, network connection, availfable devices.

- **Knowledge Gap:** Lack of formal botanical or pathological training.
- **Budget:** Cannot afford to hire professional consultants for every sick plant.
- **Time:** Diseases spread rapidly; waiting days for an expert diagnosis results in dead crops.

5. AVAILABLE SOLUTIONS AS

Which solutions are available to the customers when they face the problam.

- ***Hiring Agronomists/Experts***: (Pros: Highly accurate. Cons: Expensive, slow to schedule).
- ***Internet Search/Forums***: (Pros: Free. Cons: Time-consuming, highly unreliable, conflicting advice).
- ***Applying generic pesticides blindly***: (Pros: Fast. Cons: Harms the environment, wastes money, often ineffective against the specific disease).

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS J&P

Wlrich jobs-to-be-done (or problems) do you address for your customers? There could be more than one, explore different sides.

- **Job:** Accurately and quickly identify plant diseases from visual symptoms.
- **Problem:** Prevent crop yield loss, avoid financial damage, and stop the spread of infection to neighboring plants.
- **Problem:** Avoid guessing which pesticide/treatment to apply.

9. PROBLEM ROOT CAUSE RC

What is the real reason that this problem exists?
What is the back story behmd the need to do

- Plant diseases are visually complex, and many distinct diseases share overlapping visual symptoms (like yellowing leaves).
- There is a severe global shortage of accessible agricultural experts compared to the vast number of farmers and gardeners.

7. BEHAVIOUR BE

What doee your customer do to address the problem and get the job dong?

- Taking pictures of the sick leaves to show to others.
- Asking neighbors, local farmers, or nursery staff for their opinions.
- Pulling up diseased plants entirely to prevent spread (sometimes unnecessarily).

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS TR

What triggers customers to act? i.e. seeing their neighbour installing

- Noticing unexpected visual changes on plants (spots, yellowing, wilting, mold).
- Experiencing a sudden drop in crop yield or plant health.
- Hearing news about a local disease outbreak (e.g., tomato blight spreading in the region).

4. EMOTIONS: BEFORE / AFTER EM

How do customers feel when they face a problem or a job and afterwards?
i.e. lost, insecure - consident, in control - use it in your communication strategy & design.

Before: Anxious, frustrated, helpless, worried about financial loss or plant death.

After (Using PlantCare AI): Relieved, confident, in control, and empowered to take immediate correct action.

10. YOUR SOLUTION SL

If you are working on an exicting business, write down your current solution first, fill in the canves, and check how much it fitx reality.

- **PlantCare AI:** A mobile-responsive web application powered by a lightweight deep learning model (MobileNetV2). It allows users to instantly upload a photo of a plant leaf and receives a real-time, highly accurate classificatcn across 38 distinct disease classes, bridging the expert knowledge gap.

8. CHANNELS of BEHAVIOUR CH

8.1 ONLINE

What kind of actions do customers take offline? Extract online channels from #7

* Posting photos to Reddit (e.g., r/plantclinic), gardening Facebook groups, or Googling visual symptoms ("brown spots on tomato leaf").

8.2 OFFLINE

* Taking physical leaf samples to local garden centers or calling agricultural extension offices for help.

Identify strong TR & EM